Chapter 1 Java Language Overview

Array and Function

Software College of NEU



Array



- An array is a simple type of ordered list.
- Group data objects of the same type
- Declare arrays of primitive or class types

```
char s[];

Point p[];

char [] s;

Point [] p;
```

- Create space for a reference
- Remember an array is an object not memory reserved for primitive types

Creating Array



- Use the **new** keyword to create an array object.

```
s = new char[20];

p = new Point[100];

p[0] = new Point();

p[1] = new Point();
```

Initializing Arrays

```
1 String names[];
2 names = new String[3];
3 names[0] = "Georgianna";
4 names[1] = "Jen";
5 names[2] = "Simon";
6
7 Myclass array[] = {
8 new Myclass(),
9 new Myclass(),
10 new Myclass()
11 };
12
13 Color palette[] = {
14 Color.blue,
15 Color.red,
16 Color.white
17 };
```

Array Bounds



- All array subscripts begin at **0**:

```
int list[] = new int [10];

for (int i = 0; i < list.length; i++)
  {
     System.out.println(list[i]);
}
```

Array Resizing



- Cannot resize an array
- Can use the same reference variable to refer to an entirely new array:

```
int elements[] = new int[6];

elements = new int[10];
```

Copying Arrays

```
1 //original array
2 int elements[] = { 1, 2, 3, 4, 5, 6 };
3
4 // new larger array
5 int hold[] = { 10, 9, 8, 7, 6, 5, 4, 3, 2,
  1 };
6
7 for(int i = 0; i < elements.length; i++){
8         elements[i] = hold[i];
9 }
```

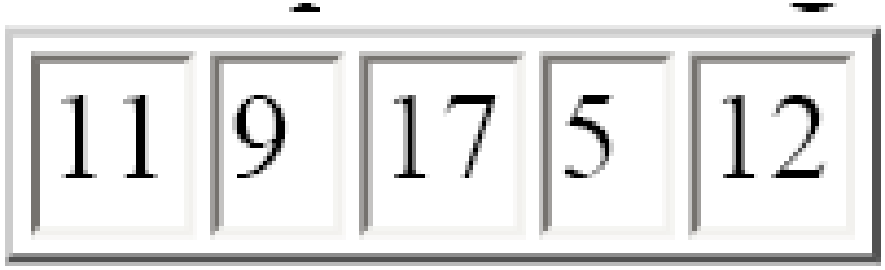
Methods



- Methods have a *signature* and *a body* of statements (in braces)
- Can have *parameters*
- Can have *a return value*
- Can have *local variables*

Selection Sort

- Sorts an array by repeatedly finding the smallest element of the unsorted tail region and moving it to the front
- Slow when run on large data sets
- Example: sorting an array of integers



Sorting an Array of Integers

- Find the smallest and swap it with the first element



- Find the next smallest. It is already in the correct place



- Find the next smallest and swap it with first element of unsorted portion



Continued

Sorting an Array of Integers

- Repeat



- When the unsorted portion is of length 1, we are done



Search method

```
public static int search(int [] a, int key) {
        for(int i = 0; i < a.length; i++) {
                if(a[i] == key) {
                        return i;
                }
        }
        return -1;
}
```

Sort method

```
public static void sort(int [ ] a) {

        int temp;
        for(int i = 0; i < a.length; i++) {
                for(int j = a.length -1; j > i; j--) {
                        if(a[j] < a[j-1]) {
                                temp = a[j];
                                a[j] = a[j-1];
                                a[j-1] = temp;
                        }

                }
        }
}
```

Main method

```
public static void main(String[] args) {

        int list[] = new int[5];
        list[0] = 98;
        list[1] = 104;
        list[2] = 389;
        list[3] = 11;
        list[4] = 9;

        sort(list);
        for(int i = 0; i < 5; i++) {
                System.out.println(list[i]);
        }

        int position;
        position = search(list, 76);
        System.out.println("The position is " + position);
}
```

Homework



- **Check the functions and discuss with your friends if they are defined properly.**

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